Java Introduction - Questions & Answers

1 Mark Questions

1. Who created Java?

- Java was created by James Gosling, Patrick Naughton, Chris Warth, Ed Frank, and Mike Sheridan at Sun Microsystems in 1991.

2. What was Java initially called?

- Java was initially called "Oak'.

3. What is JVM?

- JVM (Java Virtual Machine) is a runtime environment that executes Java bytecode.

4. What is the full form of JDK?

- JDK stands for Java Development Kit.

5. What are Java tokens?

- Java tokens are the smallest units in a Java program, including keywords, identifiers, literals, operators, and separators.

2 Mark Questions

1. Why was Java developed?

- Java was developed to create a platform-independent language for consumer electronic devices, overcoming the need for expensive and time-consuming C/C++ compilers.

2. What is the difference between JRE and JDK?

- JRE (Java Runtime Environment) provides the runtime for Java programs, while JDK (Java Development Kit) includes JRE along with development tools.

3. How does Java ensure security?

- Java ensures security by running programs inside the JVM and checking bytecode for malicious content.

4. Explain Java's platform independence.

- Java code is compiled into bytecode, which can run on any operating system with a compatible JVM.

3 Mark Questions

1. Explain the structure of a Java program.

- A Java program consists of:

1. Documentation section - Comments with program details.
2. Package statement - Defines the package.
3. Import statements - Includes external classes.
4. Class definitions - Defines one or more classes.
5. Main method - The entry point of the program.

2. What are the main features of Java?

- Simple: Syntax similar to C/C++, but no pointers or multiple inheritance.
- Object-Oriented: Uses encapsulation, inheritance, and polymorphism.
- Platform-Independent: Runs on any OS with JVM.
- Secure: No explicit pointers, bytecode verification.
- Multithreading: Supports concurrent execution of multiple tasks.

3. Differentiate between implicit and explicit type casting in Java.

- Implicit (Widening Conversion): Automatically converts smaller data types to larger ones (e.g., int to double).
- Explicit (Narrowing Conversion): Requires explicit casting when converting a larger data type to a smaller one, possibly leading to data loss (e.g., double to int).

5 Mark Questions

1. Compare C++ and Java in terms of their features.

- Pointers: C++ uses pointers, Java does not (for security).

- Memory Management: C++ requires manual memory management, while Java has automatic garbage collection.

- Multiple Inheritance: C++ supports multiple inheritance, Java uses interfaces.

- Platform Dependency: C++ is platform-dependent, Java is platform-independent via bytecode.

- Compilation & Execution: C++ compiles to machine code, Java compiles to bytecode executed by JVM.

2. Explain Java's decision-making statements with examples.

- if statement:

```
if (age > 18) {
    System.out.println("Eligible to vote.");
}
```

- if-else statement:

```
if (number % 2 == 0) {
    System.out.println("Even number");
} else {
    System.out.println("Odd number");
}
```

- Switch statement:

```
switch (day) {
    case 1: System.out.println("Monday"); break;
    case 2: System.out.println("Tuesday"); break;
    default: System.out.println("Invalid day");
}
```